

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 20

TOTAL MARKS: 20

Annexure A
MOCK INTERVIEW

Code of Conduct:

*I **Kartheeka Repalle(192372289)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Kartheeka Repalle

Annexure A
MOCK INTERVIEW

1. **Analyze the weaknesses of SSL and explain why TLS is considered more secure in modern web systems.**
2. **Apply the TLS handshake process to explain how secure session keys are established between a client and server.**
3. **Compare SSL and TLS in terms of encryption mechanisms and identify key improvements in TLS.**
4. **Evaluate a scenario where improper certificate validation in TLS leads to a security breach. What could go wrong?**
5. **Analyze the role of digital certificates in SSL/TLS and explain how they ensure authentication.**
6. **Apply your understanding of SET protocol to explain how it protects credit card transactions in e-commerce.**
7. **Evaluate the limitations of the SET protocol in current online payment systems.**
8. **Compare SET protocol with TLS-based payment security and justify which is more practical today.**
9. **Analyze a malware attack scenario and differentiate whether it is caused by a virus, worm, or Trojan.**
10. **Apply appropriate mitigation techniques for a worm attack spreading across a corporate network.**
11. **Evaluate the impact of Trojan attacks on system confidentiality and integrity with an example scenario.**
12. **Compare viruses, worms, and Trojans based on propagation method and system impact.**
13. **Analyze the working of a packet-filtering firewall and identify its limitations in modern networks.**
14. **Apply firewall rules to block unauthorized access while allowing legitimate traffic in a given scenario.**
15. **Evaluate the effectiveness of stateful inspection firewalls over stateless firewalls.**
16. **Compare different types of firewalls and recommend the most suitable one for an enterprise network.**

- 17. Analyze how an IDS detects malicious activities using signature-based and anomaly-based techniques.**
- 18. Apply IDS/IPS mechanisms to prevent a distributed denial-of-service (DDoS) attack.**
- 19. Evaluate the differences between IDS and IPS in terms of response to detected threats.**
- 20. Propose improvements to reduce false positives in an IDS while maintaining high detection accuracy.**

1. SSL VS TLS

SSL has several weaknesses, including outdated cryptographic algorithms and protocol vulnerabilities.

TLS is the modern replacement with stronger encryption, better authentication, and improved security mechanisms.

2. TLS HANDSHAKE

The client and server negotiate security parameters and authenticate the server using its certificate.

They establish shared secret key material, which is then used to encrypt the communication session.

3. SSL VS TLS ENCRYPTION

TLS improves SSL by supporting stronger cryptographic algorithms, better key exchange, and stronger authentication.

Modern TLS versions also remove many insecure and outdated features of SSL.

4. IMPROPER CERTIFICATE VALIDATION

If a client does not properly validate a server certificate, an attacker may impersonate the legitimate server.

This can enable a man-in-the-middle attack and expose sensitive information.

5. DIGITAL CERTIFICATES

Digital certificates bind a server's identity to its public key and are issued by trusted Certificate Authorities.

The client validates the certificate before trusting the server.

6. SET PROTOCOL

SET (Secure Electronic Transaction) protects card payments using encryption, digital certificates, and digital signatures.

It helps authenticate the participants and protect payment information during transactions.

7. LIMITATIONS OF SET

SET is complex and requires certificate management for participants, making deployment difficult.

Its complexity and infrastructure requirements caused modern payment systems to prefer simpler approaches.

8. SET VS TLS

SET provides strong payment-specific security but is complex to deploy and manage.

TLS is more practical today because it securely protects communication and integrates easily with web applications and payment gateways.

9. VIRUS, WORM, OR TROJAN

A virus attaches itself to files and spreads when infected files are executed.

A worm spreads automatically across networks, while a Trojan disguises itself as legitimate software and relies on user execution.

10. WORM MITIGATION

Isolate infected systems, block unnecessary network connections, and apply security patches immediately.

Use endpoint protection, network segmentation, and continuous monitoring to prevent further spread.

11. TROJAN IMPACT

A Trojan can provide unauthorized access to a system and allow attackers to steal or modify sensitive information.

For example, a fake application may secretly install malware that steals user credentials.

12. VIRUS VS WORM VS TROJAN

A virus spreads by infecting files, a worm spreads automatically through networks, and a

Trojan relies on deception.

All can damage confidentiality, integrity, or availability, but their propagation methods differ.

13. PACKET-FILTERING FIREWALL

A packet-filtering firewall checks packet information such as source IP, destination IP, port, and protocol against rules.

Its limitation is that it has limited awareness of application behavior and connection state.

14. FIREWALL RULES

Allow only required services and trusted source addresses, while blocking unnecessary ports and unauthorized traffic.

For example, allow HTTPS on port 443 and block unnecessary external access to administrative ports.

15. STATEFUL VS STATELESS FIREWALL

A stateful firewall tracks active connections and can distinguish legitimate response traffic from unsolicited traffic.

Therefore, it generally provides better security and flexibility than a stateless firewall.

16. TYPES OF FIREWALLS

Common types include packet-filtering, stateful, proxy, and next-generation firewalls.

For an enterprise, a next-generation firewall is generally preferred because it provides application awareness, advanced filtering, and threat prevention.

17. IDS DETECTION

A signature-based IDS detects known attacks by comparing traffic with predefined attack patterns.

An anomaly-based IDS compares activity with normal behavior and can identify previously unknown threats.

18. IDS/IPS AGAINST DDoS

IDS can detect unusual traffic volumes and generate alerts, while IPS can automatically block or rate-limit suspicious traffic.

Additional controls such as traffic filtering, rate limiting, and upstream DDoS protection are also useful.

19. IDS VS IPS

An IDS mainly detects suspicious activity and generates alerts without directly blocking the traffic.

An IPS works inline and can automatically block or prevent detected malicious activity.

20. REDUCING IDS FALSE POSITIVES

Use accurate detection rules, tune thresholds, maintain updated signatures, and establish a reliable normal-traffic baseline.

Regularly review alerts and use multiple indicators before classifying activity as malicious.